# FinFusionNet

## A Multimodal Framework for Short-Term Stock Direction and Volatility Prediction in the Indian Stock Market

**Undergraduate Thesis Research Proposal — Enhanced Final Draft (v2)**

Target Focus: NIFTY 50 Index | Empirical Multimodal AI | Controlled Ablation Design

**Plain-English Executive Summary: What is this project about?**

Financial markets move based on both hard numerical data (stock prices, trading volume) and soft textual data (news headlines, corporate announcements). **FinFusionNet** explores whether teaching a modern AI language model (*FinBERT*) to read Indian financial news helps predict tomorrow's stock movements better than relying on stock charts alone.

To ensure the findings are honest and scientifically sound, this study systematically turns features on and off (a process called an **ablation study**) to verify if any accuracy gains come specifically from the AI news model or just from lucky market noise.

**Key Concept Primer (For Non-Experts)**

- **Multimodal Framework:** An AI system that processes two or more distinct types of information simultaneously—here, numerical market charts + human language news.
- **FinBERT:** A specialized Artificial Intelligence language engine specifically trained to read and understand financial news and market terminology.
- **Ablation Study:** A controlled experiment where components of an AI system are stripped away step-by-step to test the exact value of each component.
- **Direction vs. Volatility:** *Direction* asks "Will the price go UP or DOWN tomorrow?" while *Volatility* asks "How violently will the price MOVE tomorrow?"
- **Temporal Cross-Attention:** A smart blending mechanism that aligns news sentiment from specific past days with corresponding technical indicators in time.

## 1. Abstract

Financial markets are shaped not only by historical price behaviour but also by the language used in news coverage. This proposal investigates whether financial language representations extracted using FinBERT improve short-term prediction of stock direction and volatility in the Indian stock market, relative to price-only baselines and simpler lexicon-based sentiment methods, and whether temporal cross-attention offers a meaningfully better fusion strategy than simple feature concatenation.

The study is structured as a controlled five-stage ablation across NIFTY 50 constituents, evaluated with walk-forward validation and tested for statistical significance using the Diebold-Mariano test. Beyond the core ablation, the study reports results separately for direction and volatility (an angle underexplored in prior FinBERT-fusion work), breaks results down by

sector, and tests robustness across pre- and post-2020 market regimes. The intended contribution is a rigorous, transparent, and reproducible empirical answer to a precisely scoped question, positioned explicitly against existing literature on FinBERT-based and cross-modal financial forecasting.

## 2. Research Question

Can financial language representations extracted using FinBERT improve short-term stock direction and volatility prediction in the Indian stock market compared to:

1. Traditional price-only models?
2. Lexicon-based sentiment baselines?

Furthermore, does **temporal cross-attention** improve multimodal fusion compared to simple feature fusion?

**Sub-Questions:**

- Does this predictive effect differ between *direction prediction* (up/down) and *volatility prediction* (degree of variation)?
- Does the predictive performance hold consistently across different market sectors (e.g., Banking vs. IT) and across macro market regimes (pre-2020 vs. post-2020)?

## 3. Background and Positioning in Existing Literature

FinBERT-based sentiment features have previously been applied to Indian equity data, including studies combining NIFTY 50 price series with large volumes of financial news processed through a fine-tuned FinBERT model. Prior work has also shown that FinBERT-derived sentiment tends to outperform traditional and lexicon-based sentiment baselines on classification accuracy, Area Under the Curve (AUC), and simulated trading performance in international markets. Separately, cross-modal and temporal attention mechanisms for fusing sentiment embeddings with market indicators represent an active research direction, with dedicated architectures proposed for cross-modal temporal fusion in financial forecasting.

**The Identified Research Gap:** What existing work has not established—particularly on Indian equity data—is a single controlled comparison that holds the prediction task, dataset, and evaluation protocol constant while isolating three specific factors:

1. The presence of any news signal at all.
2. The use of a domain-specific transformer (FinBERT) instead of a simple lexicon-based sentiment score (e.g., VADER/TextBlob).
3. The use of temporal cross-attention instead of simple feature concatenation.

In addition, most prior FinBERT-fusion studies evaluate direction prediction only; volatility prediction under the same fusion framework remains comparatively unexamined. This project is scoped to close both gaps: it runs the full controlled ablation on NIFTY 50, and it evaluates direction and volatility as two distinct, separately reported tasks.

## 4. Research Objectives

To address the research gap, this project establishes seven core objectives:

1. **Baseline Construction:** Develop robust baseline prediction models using only historical numerical market data (XGBoost, LSTM).
2. **Lexicon Signal Testing:** Test whether a traditional lexicon-based sentiment score (VADER / TextBlob polarity) improves predictive performance over price-only baselines.
3. **FinBERT Representation Testing:** Investigate whether FinBERT-derived sentiment representations improve performance beyond lexicon-based sentiment.
4. **Fusion Architecture Evaluation:** Evaluate whether temporal cross-attention provides better multimodal fusion than simple feature concatenation.
5. **Task Differentiation:** Determine whether the above effects differ between direction prediction (classification) and volatility prediction (regression).
6. **Robustness Verification:** Determine whether findings hold consistently across sectors and across pre-/post-2020 market regimes.
7. **Rigorous Protocol:** Compare all approaches using a consistent, reproducible, and statistically rigorous evaluation protocol.

## 5. Scope

- **Market Universe:** Indian Stock Market, focused on NIFTY 50 constituents, categorized by sector tags (Banking, IT, FMCG, Auto, Pharma, Energy, etc.).
- **Prediction Tasks:** Next-day direction (binary classification: $+1$ up, $0$ down) and next-day realized volatility (continuous regression), reported and analyzed separately.
- **Inputs:** Historical price/volume (OHLCV), technical indicators (RSI, MACD, EMA, ATR, Bollinger Bands), and daily financial news headlines.
- **Validation Protocol:** Strict **walk-forward validation** (rolling time windows). No random train/test splits or k-fold shuffling to prevent look-ahead bias and data leakage.
- **Regime Robustness:** Primary run on complete dataset, accompanied by a secondary regime split comparing pre-2020 (steady growth) vs. post-2020 (post-pandemic high volatility) periods.
- **Dataset Window:** Minimum 5–7 years of daily OHLCV and matched news coverage.

## 6. Data Sources

The research utilizes publicly available and reliable financial and news data infrastructure:

Table 1: Summary of Data Components and Primary Sources

| Component | Source | Technical Implementation & Notes |
|---|---|---|
| **OHLCV Price Data** | Yahoo Finance (`yfinance`) | `TICKER.NS` format; free, multi-year daily history covering Open, High, Low, Close, Volume. |
| **Technical Indicators** | `pandas-ta` (computed) | Systematically derived indicators: RSI, MACD, EMA, ATR, and Bollinger Bands from raw OHLCV. |
| **News Headlines** | GDELT Project | Open-access, multi-year depth; filtered for Indian company mentions and ticker keywords. |
| **News (Fallback/Upgrade)** | EOD Historical Data | Premium fallback used only if GDELT coverage or filtering proves insufficient for specific tickers. |

## 7. News Data Protocol

To maintain scientific validity and prevent **information leakage** (allowing future news to corrupt past predictions), a strict, automated protocol governs news alignment:

1. **Market Close Cutoff (3:30 PM IST):** Only news published *before or at* 3:30 PM IST on Trading Day $T$ is assigned to predict Trading Day $T + 1$.
2. **Post-Close Roll Forward:** Any news published *after* 3:30 PM IST on Day $T$ is automatically shifted to count toward Trading Day $T + 1$'s prediction for Day $T + 2$.
3. **Deduplication:** Duplicate and near-duplicate syndicated headlines are filtered before embedding generation.
4. **Timezone Normalization:** All source timestamps are standardized to Indian Standard Time (IST).
5. **Zero-News Fallback Rule:** Trading days without qualifying news receive a pre-defined neutral fallback vector (**0**-vector embedding / $0.0$ neutral polarity score) applied uniformly across all models.

## 8. Experimental Roadmap

The experimental roadmap follows a controlled ablation design. Each subsequent stage introduces **exactly one new variable** relative to the prior baseline, making performance attribution unambiguous.

1. **Stage 1 (Baseline A):** Historical Price Data $\rightarrow$ XGBoost Model.
2. **Stage 2 (Baseline B):** Historical Price Data $\rightarrow$ Long Short-Term Memory (LSTM) Model.
3. **Stage 3 (Lexicon Fusion):** Historical Price Data + VADER/TextBlob Polarity $\rightarrow$ Simple Concatenation Fusion.

4. **Stage 4 (FinBERT Fusion):** Historical Price Data + FinBERT Embeddings → Simple Concatenation Fusion.
5. **Stage 5 (FinFusionNet):** Historical Price Data + FinBERT Embeddings → **Temporal Cross-Attention Fusion**.

## 9. Ablation Study Design

The table below illustrates the isolated factor matrix across all 5 experimental configurations:

Table 2: Systematic Component Ablation Matrix

| Model Architecture | Price Data | News Signal | Cross-Attention | Primary Purpose / Isolated Variable |
|---|---|---|---|---|
| **1. XGBoost** | ✓ | × | × | Standard tabular gradient boosting baseline. |
| **2. LSTM** | ✓ | × | × | Sequential deep learning baseline. |
| **3. Lexicon + Fusion** | ✓ | ✓ (VADER) | × | Isolates impact of *any* basic news sentiment. |
| **4. FinBERT + Fusion** | ✓ | ✓ (FinBERT) | × | Isolates value of *transformer language models*. |
| **5. FinFusionNet** | ✓ | ✓ (FinBERT) | ✓ | Isolates value of *temporal cross-attention fusion*. |

**How to Read the Ablation Table**

- **Row 1–2 to Row 3:** Jump measures whether **adding news text** helps at all.
- **Row 3 to Row 4:** Jump measures whether **FinBERT AI** is better than simple dictionary word counts.
- **Row 4 to Row 5:** Jump measures whether **Cross-Attention Fusion** beats basic feature stacking.

## 10. Evaluation Protocol

### 10.1 Classification Metrics (Direction Prediction)

Next-day direction prediction is evaluated using five complementary classification metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

In addition, **F1-Score** (harmonic mean of Precision and Recall) and **ROC-AUC** (Receiver Operating Characteristic Area Under Curve) are reported to account for class balance.

### 10.2 Regression Metrics (Volatility Prediction)

Next-day realized volatility forecasting is evaluated using standard regression error metrics:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

### 10.3 Statistical Significance: Diebold-Mariano Test

To determine whether performance improvements are statistically meaningful or merely random artifacts, pairs of model forecasts are evaluated using the **Diebold-Mariano (DM) test** on loss differentials ($d_t = e_{1,t}^2 - e_{2,t}^2$):

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})/T}} \sim \mathcal{N}(0,1)$$

All per-fold forecast errors are logged to compute exact $p$-values. Note that the DM test is conducted in two separate runs corresponding to each task type using its native error: for classification (direction prediction), DM is applied to the squared or binary error differential based on F1/AUC loss; for regression (volatility prediction), DM is applied to the squared error differential based on RMSE. These are two separate DM test runs, not one shared formula across both tasks.

### 10.4 Confidence Intervals via Bootstrapping

Rather than reporting single point estimates, 95% bootstrap confidence intervals are computed across walk-forward folds. This provides clear error margins (e.g., Accuracy: $58.2\% \pm 1.4\%$) to substantiate comparative claims.

### 10.5 Sector and Regime Breakdown

Results are disaggregated by sector (e.g., Banking vs. IT) to identify where news sentiment offers high predictive power, and split into pre-2020 vs. post-2020 windows to test robustness across structural market shifts.

## 11. Reproducibility and Repository Structure

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To guarantee full transparency and scientific auditability, the project is engineered around a modular, version-controlled research repository structure:

```
FinFusionNet/
|-- docs/                       # Thesis proposal, literature review, methodology
|-- src/                        # Modular production code (no monolithic notebooks)
|   |-- data/                   # Data collection, GDELT scrapers, YFinance loaders
|   |-- features/               # Technical indicators & FinBERT embedding pipelines
|   |-- models/                 # XGBoost, LSTM, and FinFusionNet implementations
|   `-- evaluation/             # Walk-forward splitter, DM test, bootstrap metrics
|-- experiments/
|   |-- configs/                # Reproducible YAML configs for Experiments 1 to 5
|   `-- results/                # Logged per-fold predictions & error matrices
|-- tests/                      # Unit tests for time-series alignment & indicator math
|-- run_experiment.py           # Unified single CLI entrypoint
`-- requirements.txt            # Environment dependencies pinned with exact versions
```

Key reproducibility practices include:

- **Single Command Execution:** `python run_experiment.py --config configs/exp5_finfusionnet.yaml`
- **Programmatic Tables:** README and paper tables are dynamically compiled directly from raw execution logs in `experiments/results/`.

## 12. Research Contribution

The primary contribution of this undergraduate thesis is **not** the claim of creating an unbeatably profitable trading model. Rather, it is providing a **controlled, highly rigorous, and transparent empirical evaluation** of whether financial text embeddings and cross-attention mechanisms deliver statistically verifiable improvements in stock prediction on Indian market data.

A well-executed, honest negative or neutral result (e.g., proving that news embeddings do not overcome noise in short-term daily volatility) is treated as a completely valid, valuable, and publishable scientific outcome.

## 13. Out of Scope (Version 1)

To prevent scope creep and maintain strict academic focus, the following elements are intentionally excluded from Version 1:

- Large Generative LLMs (e.g., LLaMA-Financial, GPT-4 prompt pipelines).
- Reinforcement Learning (RL) based automated execution agents.
- Options trading strategy modeling or complex derivative pricing.
- Portfolio optimization and asset allocation frameworks.
- Cryptocurrencies and international equity indices (non-Indian markets).

## 14. Anticipated Limitations

The thesis explicitly acknowledges and analyzes the following realistic constraints:

1. **News Coverage Asymmetry:** Mid-cap or less popular NIFTY constituents may have sparse news coverage compared to heavily reported mega-caps (e.g., Reliance or HDFC Bank).
2. **Sample Size Limits:** Walk-forward validation across a 5–7 year window yields a finite number of test folds, requiring careful hyperparameter tuning to avoid overfitting cross-attention weights.
3. **Cutoff Sensitivity:** Model performance may exhibit sensitivity to the choice of the 3:30 PM IST news cutoff threshold.
4. **Noise in Global News Aggregators:** GDELT keyword filters may occasionally introduce non-price-sensitive peripheral news.

## 15. Success Criteria

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The research project will be judged successful if it yields empirical, reproducible answers to seven core questions:

1. Does lexicon sentiment beat price-only baselines?
2. Does FinBERT outperform traditional lexicon scores?
3. Does temporal cross-attention improve upon simple concatenation fusion?
4. Do predictive effects differ between stock direction and stock volatility?
5. Are findings consistent across industry sectors and market regimes?
6. Are performance variations statistically significant under the Diebold-Mariano test?
7. Are failure modes and limitations transparently documented?

## 16. Recommended Workflow

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The project execution follows a structured 9-step timeline:

## 17. Final Note

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This specification is designed to be ambitious yet fully realistic within an undergraduate research timeline. By framing the study around empirical transparency, rigorous statistical testing, and reproducibility rather than exaggerated market claims, the resulting thesis provides a strong, defensible, and academically sound foundation for financial AI research.

**9-Step Master Project Execution Plan**

| Step | Execution Phase & Key Deliverables |
|---|---|
| **Step 1** | **Literature & Positioning:** Finalize related work review; position explicitly against prior studies. |
| **Step 2** | **Dataset Design:** Finalize 5–7 year date window, NIFTY 50 universe, sector mappings, and GDELT pipeline. |
| **Step 3** | **Repository Scaffold:** Build modular `src/`, configuration YAML files, and test suites. |
| **Step 4** | **Baselines (Exp 1–2):** Implement and validate pure price XGBoost and LSTM models. |
| **Step 5** | **Lexicon Baseline (Exp 3):** Integrate VADER / TextBlob sentiment pipeline with simple fusion. |
| **Step 6** | **FinBERT Pipeline (Exp 4):** Extract FinBERT embeddings and implement simple concatenation fusion. |
| **Step 7** | **FinFusionNet Architecture (Exp 5):** Build and train temporal cross-attention model. |
| **Step 8** | **Statistical Evaluation:** Run walk-forward splits, sector/regime analysis, bootstrap CIs, and Diebold-Mariano tests. |
| **Step 9** | **Thesis Writing & Submission:** Synthesize empirical findings into thesis paper; prepare code repository for public release. |